Testarea unitară în Python

Audio Track Renamer

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Introducere

Aplicația este un utilitar Python care redenumește automat fișiere audio, după formatul `NN - Titlu.ext`.

Pentru ilustrarea strategiilor de testare am ales clasa TrackNumberParser din modulul track_number_parser.py. Aceasta este responsabilă de parsarea, validarea și normalizarea numerelor de track ale fișierelor audio și conține o singură metodă statică: validate_and_normalize().e(), format_track(), pad_track().

Configurație software:
- Python 3.12.3
- pytest 9.0.3
- mutagen 1.47.0

Configuratie hardware:
-proiectul a rulat nativ pe 3 sisteme diferite

Strategii de testare

În prima etapă, am aplicat Testarea bazată pe specificații (Black-Box Testing).

01 Partiționarea în clase de echivalență

Am împărțit datele de intrare în clase valide și invalide pentru a reduce redundanța testelor (ex: număr simplu vs. string invalid).

02 Analiza valorilor de frontieră

Am testat limitele exacte dintre aceste clase, deorece acolo apar majoritatea defectelor.

Partiționarea în Clase de Echivalență si Valori de Frontieră

Tabel clase de echivalență

ID	Regulă	track_num	max_tracks	Output așteptat
C1	Input absent	None	None	(None, None, "empty input")
C2	Input gol	""	None	(None, None, "empty input")
C3	Input doar spații	" "	None	(None, None, "empty input")
C4	Fara cifră în string	"abc"	None	(None, None, "parse error")
C5	Current este zero	"0"	None	(None, None, "zero not allowed")
C6	Current este zero (format cu slash)	"0/10"	None	(None, None, "zero not allowed")
C7	Total este zero	"3/0"	None	(None, None, "zero not allowed")
C8	Current depășește total	"7/3"	None	(None, None, "current exceeds total")
C9	Total depășește max_tracks	"3/15"	10	(None, None, "total exceeds max")
C10	Current depășește max_tracks	"8"	5	(None, None, "current exceeds max")
C11	Format simplu valid	"5"	None	(5, None, None)
C12	Format slash valid	"3/10"	None	(3, 10, None)
C13	Litere și cifre amestecate	"track3"	None	(3, None, None)
C14	Valid cu max_tracks respectat (cu total)	"3/10"	10	(3, 10, None)
C15	Valid cu max_tracks respectat (fără total)	"5"	10	(5, None, None)

Tabel clase de frontieră

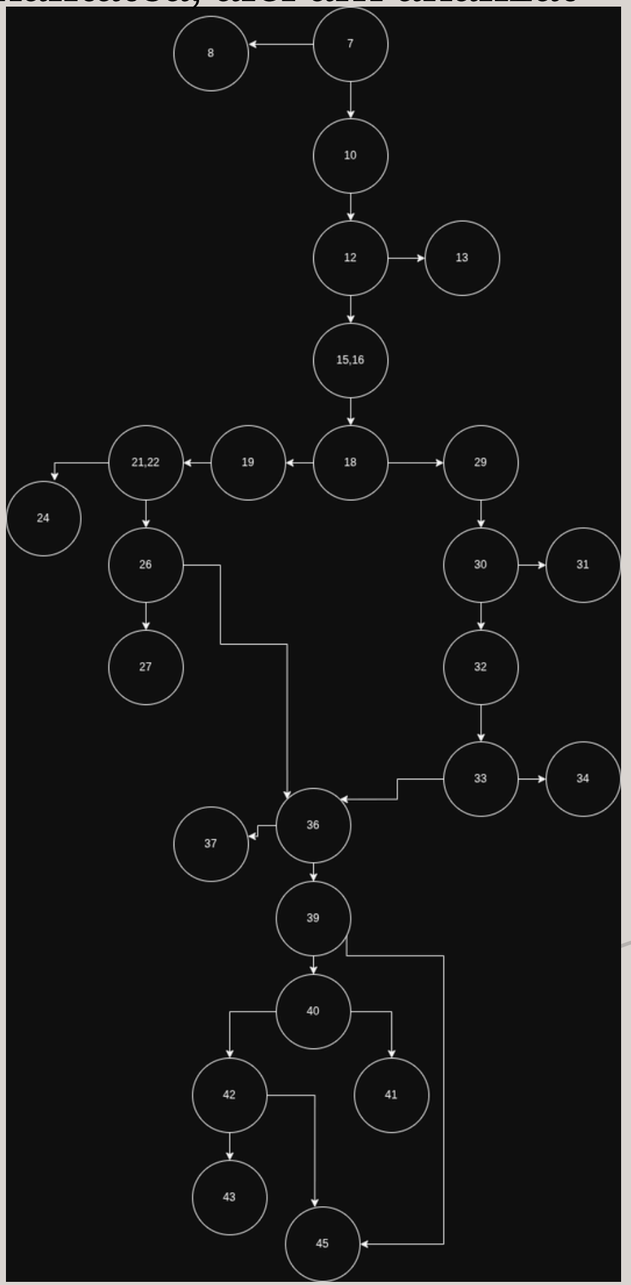
ID	Regulă	track_num	max_tracks	Output așteptat
F1	Current sub limita minimă (zero)	"0"	None	(None, None, "zero not allowed")
F2	Current exact pe limita minimă	"1"	None	(1, None, None)
F3	Current peste limita minimă	"2"	None	(2, None, None)
F4	Total sub limita minimă (zero)	"3/0"	None	(None, None, "zero not allowed")
F5	Total exact pe limita minimă	"1/1"	None	(1, 1, None)
F6	Total peste limita minimă	"1/2"	None	(1, 2, None)
F7	Current sub total cu o unitate	"4/5"	None	(4, 5, None)
F8	Current egal cu total	"5/5"	None	(5, 5, None)
F9	Current depășește total cu o unitate	"6/5"	None	(None, None, "current exceeds total")
F10	Total sub max_tracks cu o unitate	"3/9"	10	(3, 9, None)
F11	Total egal cu max_tracks	"3/10"	10	(3, 10, None)
F12	Total depășește max_tracks cu o unitate	"3/11"	10	(None, None, "total exceeds max")
F13	Current sub max_tracks cu o unitate (fără total)	"9"	10	(9, None, None)
F14	Current egal cu max_tracks	"10"	10	(10, None, None)
F15	Current depășește max_tracks cu o unitate	"11"	10	(None, None, "current exceeds max")

Toate testele unitare scrise pentru aceste clase au rulat cu succes folosind pytest.

Testare structurala

Spre deosebire de Black-Box unde am testat funcționalitatea, aici am analizat structura internă a codului.

```python
from typing import Optional, Tuple

class TrackNumberParser:

    @staticmethod
    def validate_and_normalize(track_num: Optional[str],max_tracks: Optional[int] = None) -> Tuple[Optional[int], Optional[int], Optional[str]]:
        if not track_num:
            return None, None, "empty input"

        track_str = str(track_num).strip()

        if not track_str:
            return None, None, "empty input"

        current: Optional[int] = None
        total: Optional[int] = None

        if "/" in track_str:
            parts = track_str.split("/")
            try:
                current = int("".join(c for c in parts[0] if c.isdigit()))
                total = int("".join(c for c in parts[1] if c.isdigit()))
            except (ValueError, IndexError):
                return None, None, "parse error"

            if current == 0 or total == 0:
                return None, None, "zero not allowed"
        else:
            digits = "".join(c for c in track_str if c.isdigit())
            if not digits:
                return None, None, "parse error"
            current = int(digits)
            if current == 0:
                return None, None, "zero not allowed"

        if total is not None and current > total:
            return None, None, "current exceeds total"

        if max_tracks is not None:
            if total is not None and total > max_tracks:
                return None, None, "total exceeds max"
            elif total is None and current > max_tracks:
                return None, None, "current exceeds max"

        return current, total, None
```



Acoperire la nivel de instrucțiune

Acoperire la nivel de instrucțiune (Statement Coverage)

Fiecare nod din CFG trebuie parcurs cel puțin o dată.

Test	track_num	max_tracks	Nod parcurs	Output așteptat
SC1	None	None	7, 8	(None, None, "empty input")
SC2	" "	None	7, 10, 12, 13	(None, None, "empty input")
SC3	"//"	None	7, 10, 12, 15-16, 18, 19, 21-22, 24	(None, None, "parse error")
SC4	"0/10"	None	7, 10, 12, 15-16, 18, 19, 21-22, 26, 27	(None, None, "zero not allowed")
SC5	"abc"	None	7, 10, 12, 15-16, 18, 29, 30, 31	(None, None, "parse error")
SC6	"0"	None	7, 10, 12, 15-16, 18, 29, 30, 32, 33, 34	(None, None, "zero not allowed")
SC7	"7/3"	None	7, 10, 12, 15-16, 18, 19, 21-22, 26, 36, 37	(None, None, "current exceeds total")
SC8	"3/15"	10	7, 10, 12, 15-16, 18, 19, 21-22, 26, 36, 39, 40, 41	(None, None, "total exceeds max")
SC9	"8"	5	7, 10, 12, 15-16, 18, 29, 30, 32, 33, 36, 39, 40, 42, 43	(None, None, "current exceeds max")
SC10	"3/10"	None	7, 10, 12, 15-16, 18, 19, 21-22, 26, 36, 39, 45	(3, 10, None)

Acoperire la nivel de decizie

ID	Decizie
D1	not track_num
D2	not track_str
D3	"/" in track_str
D4	except (ValueError, IndexError)
D5	cur==0 or tot==0
D6	not digits
D7	cur==0
D8	tot is not None and cur > tot
D9	max_tracks is not None
D10	tot is not None and tot > max_tracks
D11	tot is None and cur > max_tracks

Decizie	Test True	track_num	max_tracks	Output așteptat	Test False	track_num	max_tracks	Output așteptat
D1	BC1	None	None	(None, None, "empty	BC2	" "	None	(None, None, "empty
D2	BC2	" "	None	(None, None, "empty	BC3	"//"	None	(None, None, "parse error")
D3	BC3	"//"	None	(None, None, "parse error")	BC5	"abc"	None	(None, None, "parse error")
D4	BC3	"//"	None	(None, None, "parse error")	BC4	"0/10"	None	(None, None, "zero not
D5	BC4	"0/10"	None	(None, None, "zero not	BC7	"7/3"	None	(None, None, "current
D6	BC5	"abc"	None	(None, None, "parse error")	BC6	"0"	None	(None, None, "zero not
D7	BC6	"0"	None	(None, None, "zero not	BC9	"8"	5	(None, None, "current
D8	BC7	"7/3"	None	(None, None, "current	BC8	"3/15"	10	(None, None, "total
D9	BC8	"3/15"	10	(None, None, "total	BC10	"3/10"	None	(3, 10, None)
D10	BC8	"3/15"	10	(None, None, "total	BC11	"3/10"	10	(3, 10, None)
D11	BC9	"8"	5	(None, None, "current	BC11	"3/10"	10	(3, 10, None)

Acoperire la nivel de condiție (Condition Coverage)

ID	Decizie	Condiție individuală
C1	D5: cur==0 or tot==0	cur==0
C2	D5: cur==0 or tot==0	tot==0
C3	D8: tot is not None and cur > tot	tot is not None
C4	D8: tot is not None and cur > tot	cur > tot
C5	D10: tot is not None and tot > max	tot is not None
C6	D10: tot is not None and tot > max	tot > max
C7	D11: tot is None and cur > max	tot is None
C8	D11: tot is None and cur > max	cur > max

Condiție	Test True	track_num	max_tracks	Output așteptat	Test False	track_num	max_tracks	Output așteptat
C1: cur==0	BC4	"0/10"	None	(None, None, "zero not allowed")	BC7	"7/3"	None	(None, None, "current
C2: tot==0	CC1	"3/0"	None	(None, None, "zero not allowed")	BC4	"0/10"	None	(None, None, "zero not allowed")
C3: tot is not None	BC7	"7/3"	None	(None, None, "current	BC9	"8"	5	(None, None, "current
C4: cur > tot	BC7	"7/3"	None	(None, None, "current	BC8	"3/15"	10	(None, None, "total exceeds
C5: tot is not None	BC8	"3/15"	10	(None, None, "total exceeds	BC9	"8"	5	(None, None, "current
C6: tot > max	BC8	"3/15"	10	(None, None, "total exceeds	BC11	"3/10"	10	(3, 10, None)
C7: tot is None	BC9	"8"	5	(None, None, "current	BC8	"3/15"	10	(None, None, "total exceeds
C8: cur > max	BC9	"8"	5	(None, None, "current	CC2	"3"	10	(3, None, None)

Testarea circuitelor independente

n = 26 noduri
e = 38 arce
p = 1
V(G)=e−n+2p=38−26+2=14
Circuite independente:

a) 7, 8, 7
b) 7, 10, 12, 13, 7
c) 7, 10, 12, 15–16, 18, 19, 21–22, 24, 7
d) 7, 10, 12, 15–16, 18, 19, 21–22, 26, 27, 7
e) 7, 10, 12, 15–16, 18, 29, 30, 31, 7
f) 7, 10, 12, 15–16, 18, 29, 30, 32, 33, 34, 7
g) 7, 10, 12, 15–16, 18, 19, 21–22, 26, 36, 37, 7
h) 7, 10, 12, 15–16, 18, 29, 30, 32, 33, 36, 37, 7
i) 7, 10, 12, 15–16, 18, 19, 21–22, 26, 36, 39, 45, 7
j) 7, 10, 12, 15–16, 18, 29, 30, 32, 33, 36, 39, 45, 7
k) 7, 10, 12, 15–16, 18, 19, 21–22, 26, 36, 39, 40, 41, 7
l) 7, 10, 12, 15–16, 18, 29, 30, 32, 33, 36, 39, 40, 42, 43, 7
m) 7, 10, 12, 15–16, 18, 19, 21–22, 26, 36, 39, 40, 42, 45, 7
n) 7, 10, 12, 15–16, 18, 29, 30, 32, 33, 36, 39, 40, 42, 45, 7

Circuit	track_num	max_tracks	Output așteptat	Acoperit de
a) 7, 8	None	None	(None, None, "empty input")	BC1
b) 7, 10, 12, 13	" "	None	(None, None, "empty input")	BC2
c) 7, 10, 12, 15-16, 18, 19, 21-22, 24	"//"	None	(None, None, "parse error")	BC3
d) 7, 10, 12, 15-16, 18, 19, 21-22, 26, 27	"0/10"	None	(None, None, "zero not allowed")	BC4
e) 7, 10, 12, 15-16, 18, 29, 30, 31	"abc"	None	(None, None, "parse error")	BC5
f) 7, 10, 12, 15-16, 18, 29, 30, 32, 33, 34	"0"	None	(None, None, "zero not allowed")	BC6
g) 7, 10, 12, 15-16, 18, 19, 21-22, 26, 36,	"7/3"	None	(None, None, "current exceeds total")	BC7
h) 7, 10, 12, 15-16, 18, 29, 30, 32, 33, 36,	—	—	—	nefezabil
i) 7, 10, 12, 15-16, 18, 19, 21-22, 26, 36, 39,	"3/10"	None	(3, 10, None)	BC10
j) 7, 10, 12, 15-16, 18, 29, 30, 32, 33, 36, 39,	"5"	None	(5, None, None)	CI
k) 7, 10, 12, 15-16, 18, 19, 21-22, 26, 36, 39,	"3/15"	10	(None, None, "total exceeds max")	BC8
l) 7, 10, 12, 15-16, 18, 29, 30, 32, 33, 36, 39,	"8"	5	(None, None, "current exceeds max")	BC9
m) 7, 10, 12, 15-16, 18, 19, 21-22, 26, 36,	"3/10"	10	(3, 10, None)	BC11
n) 7, 10, 12, 15-16, 18, 29, 30, 32, 33, 36,	"3"	10	(3, None, None)	CC2

Testarea bazată pe mutații

```
- Mutation testing starting -

These are the steps:
1. A full test suite run will be made to make sure we
   can run the tests successfully and we know how long
   it takes (to detect infinite loops for example)
2. Mutants will be generated and checked

Results are stored in .mutmut-cache.
Print found mutants with `mutmut results`.

Legend for output:
🎉 Killed mutants.   The goal is for everything to end up in this bucket.
⏰ Timeout.          Test suite took 10 times as long as the baseline so were killed.
🤔 Suspicious.       Tests took a long time, but not long enough to be fatal.
🙁 Survived.         This means your tests need to be expanded.
🔇 Skipped.          Skipped.

1. Running tests without mutations
⠋ Running...Done

2. Checking mutants
.: 47/47  🎉 44  ⏰ 0  🤔 0  🙁 3  🔇 0
(base) carolinacostache@Carolinas-MacBook-Pro python-tss %
```

Rata de succes a testelor a fost de ~93.6% (44 din 47)